

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 21: Trigonometrical Identities

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 The three basic identities

1. Assertion (A): If $\sin^2 A + \sin A = 1$, then $\cos^4 A + \cos^2 A = 1$. Reason (R): $1 - \sin^2 A = \cos^2 A$. (a) A true, R false (b) A false, R true (c) both true, and R is the correct reason for A (d) both true, but R is not the correct reason for A **ICSE 2025, 1 mark**

SOLUTION

From the given relation, $\sin A = 1 - \sin^2 A = \cos^2 A$ (using R).

Then $\cos^4 A + \cos^2 A = \sin^2 A + \sin A = 1$. So A is true, and R is exactly what proves it.

Answer: (c)

2. Statement 1: $\sin^2 \theta + \cos^2 \theta = 1$. Statement 2: $\operatorname{cosec}^2 \theta + \cot^2 \theta = 1$. Which is valid? (a) only 1 (b) only 2 (c) both (d) neither **ICSE 2024, 1 mark**

SOLUTION

Statement 1 is a basic identity. The correct form of the second is $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$, so statement 2 is not valid.

Answer: (a)

2 Proving identities

3. Prove that $\frac{(\cot A + \tan A - 1)(\sin A + \cos A)}{\sin^3 A + \cos^3 A} = \sec A \operatorname{cosec} A$. **ICSE 2025, 4 marks**

SOLUTION

$$\cot A + \tan A - 1 = \frac{\cos^2 A + \sin^2 A - \sin A \cos A}{\sin A \cos A} = \frac{1 - \sin A \cos A}{\sin A \cos A}$$

$$\sin^3 A + \cos^3 A = (\sin A + \cos A)(\sin^2 A - \sin A \cos A + \cos^2 A) = (\sin A + \cos A)(1 - \sin A \cos A)$$

$$\text{LHS} = \frac{(1 - \sin A \cos A)(\sin A + \cos A)}{\sin A \cos A (\sin A + \cos A)(1 - \sin A \cos A)} = \frac{1}{\sin A \cos A}$$

$$= \sec A \operatorname{cosec} A = \text{RHS. Proved.}$$

4. Factorise $\sin^3 \theta + \cos^3 \theta$. Hence prove that $\frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} + \sin \theta \cos \theta = 1$. **ICSE 2024, 4 marks**

SOLUTION

Using $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$: $\sin^3 \theta + \cos^3 \theta = (\sin \theta + \cos \theta)(\sin^2 \theta - \sin \theta \cos \theta + \cos^2 \theta)$

$$= (\sin \theta + \cos \theta)(1 - \sin \theta \cos \theta)$$

$$\text{LHS} = \frac{(\sin \theta + \cos \theta)(1 - \sin \theta \cos \theta)}{\sin \theta + \cos \theta} + \sin \theta \cos \theta = 1 - \sin \theta \cos \theta + \sin \theta \cos \theta$$

$$= 1 = \text{RHS. Proved.}$$

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